MISH 16
Confined Space Entry

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DOCUMENT	MISH 16	**REVISION**	2	**EFFECTIVE**	04/07/2026
PROGRAM	MH Construction Industrial Safety & Health Program (MISH)				
LICENSES	WA: MHCONCI907R7 · OR: 765043-99 · ID: RCE-49250				

Confined Space Entry

DOCUMENT	REVISION	EFFECTIVE DATE	APPROVED BY
	2	04/07/2026	Jeremy Thamert

MH CONSTRUCTION INDUSTRIAL SAFETY AND HEALTH PROGRAM

Title

CONFINED SPACE ENTRY

Number

Revision 1 Approved By Jeremy Thamert 03/01/2025 Name Date Effective Date 3/1/2025 Page 1 of 21

1.0 PURPOSE

To define the requirements, responsibilities, and system for controlling entry and work in confined spaces, to ensure compliance with Occupational Safety and Health Administration/Washington Industrial Safety and Health Act (OSHA/WISHA) requirements.

This procedure complies with the requirements of the following:

□ OSHA 29 Code of Federal Regulations (CFR) 1910.146. □ Washington Administrative Code (WAC) 296-64-145 and 296-155-203. □ American National Standards Institute (ANSI) Z117.

2.0 RESPONSIBILITY

2.1 MH Construction Key Supervisor shall be responsible for the following:

2.1.1 Determining and designating a confined space as Permit Required or Non-Permit Required confine space.

2.1.2 Reviewing and approving confined Space Entry Permits (attachment MISH 16-A).

2.1.3 Monitoring compliance with this procedure.

2.1.4 Initiating and completing the Confined Space Entry Permit and JHA.

2.1.5 Ensuring that the Confined Space Entry Permit and JHA are posted and complied with.

2.1.6 Ensuring that the Confined Space sign is posted.

2.1.7 Ensuring that all personnel are sufficiently trained (before work is performed) in accordance with the requirements of subsection 5.15 of this procedure.

2.1.8 Ensuring that the atmospheric testing of confined spaces is performed according to the confined space Entry Permit.

2.1.9 Providing all air quality testing equipment required to monitor the confined space.

2.1.10 Ensuring that managers/supervisors in charge of confined space entry operations maintain a working knowledge of this procedure. (See also special requirements for “entry supervisors” at subsections 5.3 and 5.15.)

2.2 The entry supervisor shall be responsible for compliance with this procedure and for the duties specified at subsection 5.3

2.3 The standby attendant shall be responsible for compliance with this procedure and for the duties specified at subsection 5.4.

2.4 Authorized entrants shall be responsible for compliance with this procedure and for the duties specified at subsection 5.5.

2.5 Rescue service must be on-site for immediately dangerous to life and health (IDLH) conditions while work is being performed.

2.6 A single annual review, including all confined spaced entries during the previous 12-month period, should be completed. If no entry is performed during a 12-month period, no review is necessary.

3.0 DEFINITIONS

Authorized Entrant: Any individual who is authorized to enter a confined space by merit of having successfully completed all required confined space entry training.

Blanking: The absolute closure of a pipe, line, or duct to prevent passage of any material, by fastening a solid plate or “cap” across the pipe, etc.

Competent Person: An individual who is knowledgeable in confined space entry. This includes the identification of existing and potential hazards, safe means of entry and egress, criteria of the confined space permit, monitoring of atmospheric conditions, and standby attendant responsibilities; and on who has the authority to take prompt corrective action to eliminate those hazards.

Confined Space: A space which, be design, has limited means of entry and exit, may be subject to accumulations of toxic or flammable contaminants, or may be subject to an oxygen deficient/oxygen enriched atmosphere. Confined or enclosed spaces include, but are not limited to: Storage tanks, process vessels, bins, boilers, ventilation or exhaust ducts, sewers, underground utility vaults, tunnels, pipelines, and open top spaces (more than 4 feet in depth) such as pits, tubs, vaults, and vessels. Confined spaces are classified by two categories.

 Non-Permit Confine Space: An area, which meets the definition of a confined space but requires only minimal preventative measures for the protection of personnel.

 Permit Required Confine Space: A confined space, which has the potential for causing injury or illness, if preventative measures are not used. A confined space which presents a situation that is “immediately dangerous to life or health” (IDLH). These include oxygen deficient, oxygen enriched, explosive/flammable, or toxic atmospheres and areas that may become hazardous as a result of work activity.

NOTE: A CONFINED SPACE IS ALWAYS CONSIDERED TO BE IDLH UNTIL PROVEN OTHERWISE!!!

Confined Space Entry Permit: The written authorization for entry into a confined space, under established conditions, for a stated purpose during a specified work shift.

Double Block and Bleed: Isolating a confined space from a pipe by locking closed two in-line valves and by locking open a drain, or bleed, to the atmosphere in the line between the two closed valves.

Entry: Any action resulting in a part of the body breaking the plane of any opening of the confined space and including all subsequent work activities inside the confined space.

Entry Supervisor: An individual (manager, supervisor, foreman, etc.) who has been designated as being in charge of the specific confined space operation and provides the supervisory functions described in subsection 3.4.

NOTE: An entry supervisor also may serve as a standby attendant or as an authorized entrant, as long as that person is trained and equipped as required by this procedure for each role he or she fills.

Flammable Atmosphere: An atmosphere that contains a concentration of flammable vapor or gas greater than 10 percent of the lower explosion limit (LEL). (Such atmospheres may also be toxic.)

Hazardous Atmosphere: Any atmosphere containing one or more of the following:

• A potential flammable hazard.

• An oxygen deficient or oxygen enriched atmosphere.

• An atmospheric concentration of any substance listed in 29 CFR 1910, Subpart Z, at a concentration at or above the listed permissible exposure limit.

• An atmospheric concentration of any substance not listed in 29 CFR 1910, Subpart Z, but listed with a permissible exposure limit on a Material Safety Data Sheet required by 29 CFR 1910.1200, or that is otherwise known to present a safety or acute health hazard.

• Any IDLH condition.

Inerting: Rendering the atmosphere of a confined space nonflammable, non-explosive, or otherwise chemically non-reactive by displacing or diluting the original atmosphere with a gas that is non-reactive in the specific circumstance.

NOTE: Inert atmospheres are incapable of supporting human life even if they are not themselves toxic. Therefore, all inert atmospheres are IDLH.

Lower Explosive Limit (LEL): The minimum concentration of a combustible/flammable gas or vapor in air (usually expressed in percent by volume), which will ignite if an ignition source is present.

Oxygen Deficient Atmosphere: An atmosphere containing less than 19.5% oxygen by volume. (Air purifying respirators cannot be used in an oxygen deficient atmosphere.)

Oxygen Enriched Atmosphere: An atmosphere containing more than 21.0% oxygen by volume. Oxygen-rich atmospheres expand the range of flammability and will cause combustible materials to burn violently when ignited.

Retrieval Line: A line or rope attached at one end to a full body harness, chest harness, or wristlets worn by the entering employee and secured to an anchor point or lifting device located outside the entry portal.

Standby Attendant: A trained person who remains outside the defined limit of the confined space. The attendant is in communication with, or in direct view of, the person(s) inside and is trained in, and familiar with, the emergency response plans.

Toxic Atmosphere: An atmosphere in which airborne contaminants are present in concentrations above the threshold limit value/permissible exposure limit (TLV/PEL).

4.0 GENERAL REQUIREMENTS

4.1 All confined space work shall be in accordance with this procedure.

4.2 When deemed appropriate MH Construction IS&H may, after careful evaluation, grant exemption from specific requirements of this procedure for specified confined spaces, on an individual basis. Justification for granting such exemptions shall be documented on the confined space entry permit or on an attachment thereto.

4.3 Confined spaces shall be assumed to be immediately dangerous to life or health (IDLH) until proven otherwise.

4.4 All confined spaces shall be posted with the appropriate confined space sign.

4.5 A Confined space Entry Permit shall be posted at any active confined space worksite. A copy of this procedure shall be available at designated locations accessible to employees.

4.6 Atmospheric testing and continuous monitoring shall be performed in accordance with subsection 5.7.

4.7 A safe means of entry and exit (i.e., ladders, stairways, descent/retrieval devices, etc.) shall be maintained at all times.

4.8 Entry supervisors, standby attendants, and all authorized entrants shall successfully complete appropriate confined space training prior to being allowed to participate in confined space operations. (Refer to subsection 5.15.)

4.9 All other employees shall receive training to ensure that they are aware of the potential dangers of confined spaces and understand that entry is not allowed without special authorization.

4.10 Lock and tag procedures shall be in place as required by MISH 15.

4.11 The minimum level of non-task lighting for all indoor and outdoor activities shall be an average of 3 foot candles measured 30 in. above the floor or work surface.

4.12 In accordance with subsection 5.6, confined spaces shall be cleaned to the extent possible prior to personnel entering. Disposal of cleaned materials shall be in accordance with the contract environmental requirements.

4.13 Whenever a condition arises which is not planned for under the confined space permit (i.e., the air quality within a confined space deteriorates), the confined space shall be evacuated and the entry permit canceled until the situation can be resolved. An MH Construction IS& H representative shall participate in the resolution of such a situation.

NOTE: A permit is considered “canceled” when the standby attendant or entry supervisor removes the permit (and the corresponding Daily Confined Space Entry Log) from its place of posting and takes sole possession of same, thereby indicating that entry into the confined space is not allowed. The “canceled” permit may be re-issued upon resolution of the problem(s) which caused the cancellation. If the permit is permanently canceled, the work “canceled” shall be written across the face of the permit.

A permit is considered to be “expired” when its validation period is expired or the work for which the permit was issued has been completed or otherwise terminated. The word ”expired” shall then be written across the face of the permit.

4.14 When respiratory protection is required, the provisions of 29 CFR 1910.134 and WISHA Part E shall be followed. Air-line respirators used in confined spaces must be equipped with an auxiliary self-contained air supply with a minimum 5-minute capacity.

4.15 An entry supervisor responsible for the duties described in subsection 5.3 shall be assigned to each active confined space work site. The entry supervisor may also serve as a standby attendant or as an authorized entrant as long as that person is trained and equipped as required by this procedure for each role he or she performs.

4.16 A standby attendant responsible for the duties described in subsection 5.4 shall be assigned to each active confined space work site. Standby attendants shall monitor only one confined space at a time.

4.17 A confined space entry operations review system shall be administered in accordance with subsection 5.16.

5.0 PROCEDURE

5.1 INSPECTION AND EVALUATION

5.1.1 Prior to employee entry, each confined space shall be evaluated to identify the hazards and potential hazards of that space based on the following:

- The physical characteristics of the confined space
- The current and past uses of the confined space which may affect the oxygen level, toxicity, or flammability of the atmosphere in the space, or which may have left a residue of material which could fall on, engulf, or otherwise injure an employee
- Any special hazards arising from the location of the space (e.g., relative to other operations, processes, or structures) which may endanger the occupant(s) of the space.
- Any operations or processes that are to be carried out in, or which involve, the confined space while employees are inside (i.e., welding, grinding, cleaning, removal or application of paints/coatings, toxic materials, etc.).
- The frequency of routine employee entry

5.1.2 Safety Data Sheets (SDS’s) for any chemical used or encountered during confined space entry operations are made available before the entry.

5.2 CONFINED SPACE ENTRY PERMITS/LOGS

5.2.1 A Confined Space Entry Permit shall be completed prior to any employee entering a Permit Required confined space.

5.2.2 The entry permit shall authorize personnel entry into a specified confined space on a specified day and during a particular shift.

NOTE: A Confined space Entry Permit may be valid for more than one day when deemed appropriate by MH Construction IS&H (see subsection 4.2).

5.2.3 The entry permit shall include:

 A description of the hazards known or expected to be present in the confined space due to its current or past use, and any cleaning, purging or inerting performed prior to entry.

 A description of any additional hazards that may be generated as a result of activities carried out during occupancy of the space.

 A description of the type of personal protective equipment that will be required for entry.

5.2.4 The entry permit shall specify:

 The types of retrieval equipment necessary for rescue purposes that must be present at the confined space at the time of entry.

 The minimum acceptable environmental and physical condition for initial entry and occupancy of the confined space.

The pre-entry atmospheric testing to be completed immediately prior to entry.

 The type of continuous atmospheric monitoring to be conducted.

5.2.5 The entry permit shall specify all control measures that must be taken prior to entry. These measures may include but are not limited to the following:

 Procedures for isolating the space by such means as lockout/tagout of power sources, valves, or mechanical power transmission equipment; blanking, double block and bleed, or misalignment or removal of sections of pipes, ducts or lines

 Any cleaning, purging, or inerting, which can be done prior to entry into the space

 Mechanical ventilation of the space, including the capacity of blowers and how long they were run before entry

 The manner of inerting of the space if such measures are to be used

5.2.6 Authorization for welding, cutting, or heating shall be noted on the Confined Space Entry Permit upon completion of the Welding, Cutting, and Heating Permit (see MISH 14).

5.2.7 The entry permit shall receive the following approval signatures prior to issuance:

 Superintendent/manager Entry supervisor Standby attendant

5.2.8 The entry supervisor shall review the entry permit requirements with the authorized entrants and attendants before entry.

5.2.9 Only work that is originally identified on the entry permit shall be conducted in a confined space. A new permit must be obtained whenever changes in work conditions or work activities introduce new hazards into the space.

5.3 ENTRY SUPERVISOR (Competent Person)

5.3.1 The entry supervisor shall be designated as the competent person for the confined space operation. This designation shall be documented on the competent person designation (see MISH 3).

5.3.2 The entry supervisor shall perform the following functions:

 Determine if acceptable entry conditions are present at a confined space where entry is planned

 Authorize entry and oversee all confined space entry operations.

 Recognize the hazards that may be faced during entry, including information on the mode, signs or symptoms, and consequences of the exposure.

 Verify, by checking that the appropriate entries have been made on the confined space entry permit and log, that all tests specified by the permit and log have been conducted and that all procedures and equipment specified by the permit are in place before endorsing the permit and allowing entry to begin.

 Terminate the entry and cancel the permit whenever a prohibited condition is found to exist.

 Verify that rescue services are available and that the means of summoning them are operable.

 Remove unauthorized individuals who enter or who attempt to enter the confined space during entry operations.

 Determine, whenever responsibility for an entry operation is transferred and at intervals dictated by the hazards and operations performed within the confined space, that entry operations remain consistent with terms of the entry permit and that acceptable entry conditions are maintained.

 Complete all training as required by subsection 5.15.

5.4 STANDBY ATTENDANTS

5.4.1 A standby attendant, properly instructed in applicable emergency procedures, shall be present outside a confined space at all times when other personnel are entering, within, or exiting.

5.4.2 The standby attendant shall perform the following functions:

 Recognize the hazards that may be faced during entry, including information on the mode, signs or symptoms, and consequences of the exposure.

 Monitor the entry of personnel and ensure that the Daily Confined Space Entry Log is completed for each entry.

 Continuously maintain an accurate count of authorized entrants in the confined space.

• Maintain constant communication (sight, voice, radio, etc.) with those inside the confined space at all times.

 Perform pre-entry atmospheric testing and continuous atmospheric monitoring of the confined space and record the results on the Daily Confined Space Entry Log.

 Remain alert for any changes in the behavior of those inside which may signal a toxic reaction or some other serious stress reaction.

 Ensure that all required protective and emergency equipment is present and in operable condition.

☐ Complete necessary training to the extent required by his/her work activities and by this procedure.

☐ Maintain clear and safe access to and from the confined space and keep unauthorized personnel away from the immediate area.

NOTE: The standby attendant has the authority to make on-site determinations as to whether access by other personnel is appropriate and to stop work when he/she feels that such access is not appropriate.

☐ Order the authorized entrants to evacuate the confined space immediately if any of the following conditions are detected:

• A prohibited condition.

• Behavioral effects of hazard exposure in an authorized entrant.

• A situation outside the space that could endanger the authorized entrants.

• The standby attendant cannot effectively and safely perform all of his/her assigned duties as required by this procedure.

In the event of an emergency, immediately notify the Fire Department Emergency Rescue Team (911). Where a telephone is not IMMEDIATELY available to the standby attendant, other failsafe means shall be established to provide prompt communication with the Fire Department Rescue Team.

5.4.3 Standby attendants (or any other personnel) shall not enter a confined space to affect a rescue unless they are part of a trained emergency rescue team. If the standby attendant can affect a rescue without entering the confined space, he/she should do so.

5.4.4 THE STANDBY ATTENDANT SHALL PERFORM NO DUTIES OTHER THAN THOSE SPECIFICALLY IDENTIFIED IN THIS PROCEDURE.

EXCEPTION: The Standby Attendant may be assigned responsibility for monitoring the breathing air bottle cart when the cart is located within 10 ft of his/her designated location and there are no obstructions to vision or travel between that designated location and the cart. This includes VISUAL MONITORING ONLY, and does not include changing the bottles.

5.5 AUTHORIZED ENTRANTS: Authorized entrants shall perform the following functions:

☐ Complete all required training prior to entering a confined space (see subsection 5.15).

☐ Comply with all requirements specified in this procedure and the entry permit.

☐ Read the Confined Space Entry Permit and sign the log.

☐ Read the JSA and sign the Work Plan Sign-Off Log attached to the JSA.

☐ Ensure that the Daily Confined Space Entry Log is completed prior to entry.

☐ Ensure that atmospheric testing of the confined space has been performed and recorded on the log prior to signing.

☐ Know the hazards that may be faced during entry, including information on the mode, signs or symptoms, and consequences of exposure.

☐ Possess the knowledge to safely use all required equipment, including but not limited to, testing and monitoring equipment, ventilation equipment, communications equipment, PPE, lighting equipment, barriers and shields, ladders and other access/egress equipment, rescue and emergency equipment, and any other equipment necessary for the safe conduct of work in the confined space.

☐ Communicate with the standby attendant to enable the attendant to monitor entrant status and to enable the attendant to alert entrants of the need to evacuate the space as required.

• Alert the standby attendant whenever he/she recognizes any warning sign or symptom of exposure to a dangerous situation or prohibited condition.

☐ Exit from the confined space as quickly as possible whenever any of the following occur:

• An order to evacuate is given by the standby attendant or the entry supervisor.

• Any warning sign or symptom of exposure to a dangerous situation is recognized.

• A prohibited condition is detected.

• An evacuation alarm is activated.

5.6 CLEANING AND PURGING

5.6.1 Any confined space that has been used to store liquids, chemicals, or organic materials shall be purged of all residue.

5.6.2 When required, provisions for the cleaning and purging of confined spaces shall be determined on a case-by-case basis. The procedure for such cleaning and purging shall be documented on a JHA.

5.6.3 When the previous contents/use of a confined space are not known, thorough evaluation and/or testing shall be conducted to identify any potential for the presence of toxic substances, residues, or vapors.

5.7 ATMOSPHERIC TESTING

5.7.1 Atmospheric testing shall be conducted by the standby attendant, entry supervisor, or other designated individual who has received the appropriate level of training relative to the specific test equipment being used.

5.7.2 Atmospheric testing/monitoring instruments, including their power source, shall be properly maintained, set up and calibrated according to manufacturers' recommendations.

5.7.3 Prior to entry, confined spaces shall be tested for presence of oxygen-deficient, oxygen-enriched, and flammable atmospheres. The oxygen content is tested prior to or in conjunction with testing for the presence of flammable atmospheres. (Oxygen enrichment or deficiency may adversely affect the accuracy of the testing equipment when testing for LEL.)

5.7.4 Atmospheric testing of confined spaces shall include various levels of the entire portion to be occupied. Areas that cannot be tested from outside the space shall be tested as the entry progresses using extension poles or other means to test the atmosphere at least 5 ft ahead of the entrant(s).

5.7.5 Results of pre-entry testing shall be recorded on the Daily Confined space Entry Log.

5.7.6 Continuous monitoring of oxygen and LEL levels shall be conducted in all confined spaces. Alarms for oxygen deficiencies shall be set at 19.5% and oxygen-enriched atmospheres at 23% by volume. Alarms for flammable atmospheres shall be set at 10% of the LEL.

5.7.7 When CONTINUOUS MONITORING READING DEVIATE from acceptable limits, those deviations shall be explained on the Daily Confined Space Entry Log. Resolutions (corrective action) to these deviations shall also be recorded on the log.

5.7.8 If presence of toxic substances is known or suspected, testing shall be performed by the Contractor as deemed appropriate by MH Construction IS & H.

5.7.9 If presence of physical hazards (noise, irritants, corrosives, etc.) is known or suspected, testing shall be performed by the Contractor as deemed appropriate by MH Construction IS&H.

5.7.10 Employees, or their representatives, must be given an opportunity to request the space be re-evaluated.

5.8 VENTILATION

5.8.1 Mechanical ventilation shall be used whenever atmospheric hazards (e.g. toxic gases, oxygen deficiencies, etc.) are encountered.

5.8.2 If flammable gases, vapors or dusts are present, mechanical ventilation shall be of a type approved for the specific hazardous location. Such ventilation shall begin before entry and continue throughout occupancy to eliminate hazards and reduce or eliminate the need for personal protective equipment.

5.8.3 In confined spaces where there is a potential for the atmosphere to become IDLH:

 Ventilation equipment shall be equipped with and audible warning device to indicate failure of that equipment, or

 Ventilation equipment shall be positioned so that its failure is easily detectable by the standby attendant, and

 Factors such as noise, distance, configuration and proximity shall be such that the standby attendant may promptly and easily notify all authorized entrants when a failure has occurred in the system.

5.8.4 UNDER NO CIRCUMSTANCES SHALL PURE OXYGEN BE INTRODUCED INTO A CONFINED OR ENCLOSED SPACE FOR PURPOSES OF VENTILATION OR IMPROVING BREATHING AIR QUALTIY. This does not preclude the properly controlled use of oxygen/acetylene torches as described in subsection 5.9 of this procedure.

5.8.5 Local exhaust ventilation shall be provided to eliminate contaminants from welding and other operations within the confined space.

5.8.6 Continuous general ventilation (either natural or forced) shall be established and maintained during hot work.

5.8.7 Ventilation EXHAUST OUTLETS shall not be directed to locations adjacent to ventilation INTAKES, nor to locations that will expose workers.

5.8.8 In the event of a ventilation system failure, the confined space shall be evacuated immediately.

5.9 WELDING, CUTTING, AND HEATING

5.9.1 Welding, thermal cutting, and heating (hot work) shall be performed in accordance with the requirements of MISH 14. A Welding, Cutting, and Heating Permit shall be completed prior to the start of such work.

5.9.2 Welding, thermal cutting, or heating shall not be initiated in a confined space until atmospheric tests have been performed for oxygen and flammables and the results indicate an acceptable atmosphere as follows:

 Oxygen levels within the confined space shall be greater than 19/5% but less than 23.0% by volume

 Combustible/flammable gases or vapors shall be at levels below 10% of the LEL.

5.9.3 Atmospheric oxygen and LEL testing of the confined space shall be conducted CONTINUOUSLY during welding, cutting, or heating.

5.9.4 The flammable/combustible or toxic nature of any coating shall be determined before welding, cutting, or heating is initiated in a confined space. Such coatings shall be removed from the location of the welding, cutting, or heating for a distance sufficient (at least 4 in. wherever possible) to prevent ignition or out-gassing (from temperature increase) of coating materials in the unstripped areas. If sufficient stripping is not possible, respiratory protection and/or ventilation shall be provided as appropriate.

5.9.5 Whenever feasible, local exhaust shall be provided by the Contractor to capture contaminants at the source and remove them from the atmosphere in the confined space. When adequate local exhaust is not feasible, appropriate respiratory protection shall be used.

5.9.6 Gas cylinders shall be secured outside of the confined space and shall be shut off at the valves when the equipment is not in use. Cylinders containing oxygen, acetylene, or other gases shall not be taken into confined spaces.

5.9.7 Gas welding hoses, torches, and related equipment shall be carefully inspected. Hoses with leaks shall not be used in or near a confined space.

5.9.8 When left unattended for extended periods of time, gas hoses and torches shall be removed from the confined space. Gas hoses need not be removed from confined spaces in radiation zones provided the hoses are disconnected from their source.

5.9.9 Electric arc welding machines shall not be taken inside a confined space unless absolutely necessary. Electric arc welding machines may be placed within a confined space ONLY with the following assurances:

 Atmospheric testing and evaluation shall have demonstrated that there is no potential for flammable or oxygen enriched atmospheres.

 Placing the machine(s) in the confined space shall not create an electrical shock hazard.

 Access/egress shall not be obstructed.

 Ventilation shall not be adversely affected.

☐ Emergency rescue operations shall not be hindered.

5.9.10 When electrode holders are left unattended or unused for a period of time (such as a t breaks or lunch periods) the electrodes shall be removed from the holders, the holders placed in a safe location or protected, and the power switch to the equipment turned off.

5.9.11 If unattended for extended periods of time (such as overnight), electrode holders, cables, and other equipment shall be removed from the space and the power supply to the equipment disconnected. This equipment need not be removed from confined spaces in radiation zones provided the equipment is disconnected from its source.

5.9.12 Extraneous flammable or combustible materials (such as scrap wood, paper, rope, rags, etc.) shall be removed from the confined space. Combustible materials which cannot be removed shall be adequately protected.

5.9.13 Suitable fire extinguishing equipment shall be provided based on the nature and extent of flammables or combustibles present and the fires which may be expected to occur.

5.9.13.1 Carbon dioxide extinguishers may be used ONLY after it has been determined through prejob planning that discharge of carbon dioxide into the space is not likely to cause a hazard to workers.

5.9.13.2 Water extinguishers or water hoses equipped with fog nozzles or fog applicators are most suitable for hot work in the presence of ordinary (Class A) combustible materials or flammable residues or coatings.

5.10 FALL PROTECTION AND RETRIEVAL

5.10.1 Applicable fall protection requirements (including a written fall protection plan) shall be followed whenever appropriate by nature of the work environment (refer to MISH 11 and applicable OSHA/WISHA standards).

5.10.2 Before allowing entry into a confined space, the entry supervisor shall develop a plan of action for timely retrieval of individuals in the space in case of emergency and review the plan with the authorized entrants and attendants.

5.10.3 To facilitate nonentry rescue, appropriate retrieval systems or methods (as specified in the Confined Space Entry permit) shall be used whenever an authorized entrant enters a Permit Required confined space, unless the retrieval equipment would increase the overall risk of entry or would not contribute to the rescue of the entrant(s).

5.10.4 If employees are to be lowered into a confined space by a hoist, winch, or similar device, a retrieval line and body harness or a boatswain's chair shall be used. If the retrieval line and body harness are used to lower an employee to the working surface, the employee shall not detach the retrieval line from the harness.

5.10.5 If an employee is to work suspended from the hoist line in a confined space, a boatswain's chair shall be provided. The boatswain's chair shall meet the requirements of OSHA Standard 29 CFR 1910, Subpart D. The retrieval line and harness combination are not a substitute for a boatswain's chair.

NOTE: Paragraphs 5.10.4 and 5.10.5 are NOT intended to permit the use of a powered crane in conjunction with a boatswain's chair. The movement of personnel by means of a powered crane may be accomplished ONLY in accordance with MISH 21 (or in emergency rescue situations as stated in paragraph 5.10.6).

5.10.6 A crane may be utilized, as an emergency retrieval device only when careful evaluation has confirmed that such use will not expose employees to additional or greater hazards. Such use must be pre-approved by MH Construction IS&H.

5.11 FIRST AID

5.11.1 A person(s) holding a valid certificate of First Aid/CPR training shall be available at all confined space worksites, at all times when employees are present.

5.11.2 The standby attendant, foreman, or other designated individual (if certified) may serve as the on-site First Aid/CPR Attendant.

5.12 ISOLATION

5.12.1 All pipes/lines leading into or out of enclosed spaces shall be physically isolated to prevent flow or drainage into the confined space by at least one of the following means:

☐ Disconnect and cap or blank ☐ Double block and bleed ☐ Misalignment or removal of section(s)

5.12.2 When the provisions of paragraph 5.12.1 cannot be met, MH Construction IS&H shall be contacted and alternative protective measures implemented as appropriate.

5.12.3 The provisions of paragraphs 5.12.1 and 5.12.2 do not apply to pipes/lines which simply "pass through" the confined space unless the activity within the confined space may cause the pipes/lines to create a hazard (i.e., welding, cutting, or heating in the presence of lines carrying flammables, combustibles, caustics, toxics, pressurized materials, etc.).

5.12.4 Barriers are provided as necessary to protect entry personnel from external hazards.

5.13 MECHANICAL HAZARDS

5.13.1 Wherever the accidental or intentional restarting of mechanical equipment may pose a hazard to employees working within a confined space, such equipment shall be locked out and tagged or otherwise isolated in accordance with MISH 15.

5.13.2 The provisions of paragraph 5.13.1 shall also apply wherever restarting of mechanical equipment may cause ignition of flammable vapors.

5.13.3 Situations may arise where mechanical parts which are not connected to an energy source could be caused to move due to the actions of employees conducting work within (or outside of) the confined space. Such moveable parts shall be secured by means of latches, chains, chocks, blocks or other devices as appropriate.

5.14 ELECTRICAL HAZARDS

5.14.1 Electrical tools, cords, lighting, and equipment used in flammable or potentially flammable atmospheres shall be of a type approved for use in the specific hazardous location.

5.14.2 Ferrous metal tools, two-way radios, remote control units, or other static charge or spark producing equipment shall not be introduced into a Permit Required confined space unless the atmosphere is tested and monitored for LEL and proven to be non-explosive.

5.14.3 All required power tools, power cords, welding machines, and other electrical equipment shall be maintained and operated in accordance with safety standards as prescribed by OSHA/WISHA, American National Standards Institute (ANSI), National Fire Protection Association/National Electric Code (NFPA/NEC), etc.

5.14.4 In addition to the ground fault circuit interrupter (GFCI) requirements of MISH 29, power tools and electrical lighting used in wet, damp, or metal-enclosed (conductive) confined spaces shall also be protected by GFCIs.

5.14.5 Wherever the accidental or intentional re-energizing of electrical equipment of circuits may pose a hazard to employees working within a confined space, such equipment or circuits shall be locked out and tagged or otherwise isolated in accordance with MISH 15 until the work within the confined space is complete.

5.15 TRAINING

5.15.1 Formal confined space entry training shall be provided by the contractors for entry supervisors, standby attendants, and authorized entrants prior to commencement of confined space work activities. This formal training shall include instruction on:

□ Compliance with the requirements of MISH 16. □ The proper use and care of atmospheric testing equipment. □ The operation of emergency rescue/fall protection equipment.

Formal training shall be repeated every 2 years.

5.15.2 When information provided on the Confined space Entry Permit or other pre-job planning documents indicates conditions for which additional site-specific training is needed, the Contractor shall ensure that such training is provided prior to any personnel entering the confined space. Such training shall normally include the following:

□ Site-specific emergency response/rescue procedure □ Site-specific hazard recognition/avoidance training □ Site-specific personal protective equipment requirements

NOTE: The intent of site-specific training shall be to ensure that all affected employees are fully aware of the hazards and the requirements applicable to the specific confined space in which they will be working.

5.15.3 Additional training shall be provided to an employee before there is a change in his/her assigned duties.

5.15.4 Training record shall include employee name, trainer signature/initials and dates of training. Training records must be made available to employees and their authorized representative(s).

5.15.5 When multiple employers are working in the same confined space MHC will have a meeting with others to ensure that ALL employers will be adhering to the same safety rules and procedures.

5.15.6 Additional training shall be provided whenever there is reason to believe either that there are deviations from this procedure or that there are inadequacies in employee knowledge or use of this procedure.

5.15.7 Employees who have a legitimate need to enter a confined space for a brief period for inspection purposes (engineers, quality control, etc.) are not required to complete the formal confined space entry training provided:

□ They have been provided with the site-specific briefing as described in paragraph 5.15.2; and

□ They are escorted by an employee who has completed all CSE training requirements, including the formal confined space entry course; and

□ The IS&H department (or area IS&H representative) is notified prior to such an entry taking place.

6.0 RECORDS

Document

Record Submittal Responsibility Record Retention Responsibility Competent Person Designation Contractor Construction Management Confined Space Entry Permit Contractor Construction Management Daily Confined Space Entry Log Contractor Construction Management Welding, Cutting, and Heating Permit Contractor Construction Management

7.0 REFERENCES

ANSI Z117.1 Safety Requirements for Confined Spaces

8.0 ATTACHMENTS

MISH 16-A Confined Space Entry Permit

MH CONSTRUCTION CONFINED SPACE ENTRY PERMIT

Project:_____ Project #:_____ Project Location:_____ Space Description:_____ _____ Purpose of Entry:_____ _____ Permit valid from: (date)_____THRU (date)_____

HAZARD IDENTIFICATION

yes no yes no yes no ( ) ( ) oxygen deficiency(<19.5%) ( ) ( ) mechanical hazards ( ) ( ) radiation ( ) ( ) oxygen enrichment (>23.5%) ( ) ( ) electrical hazards ( ) ( ) entrapment/engulfment ( ) ( ) flammable gases/vapors ( ) ( ) dust ( ) ( ) limited egress ( ) ( ) inert atmosphere ( ) ( ) temperature extremes ( ) ( ) fall hazards ( ) ( ) welding/cutting fumes ( ) ( ) pressured fluids/gases ( ) ( ) high noise levels ( ) ( ) toxic gases or vapors (specify)_____ ( ) ( ) introduction of hazardous materials (specify)_____ ( ) ( ) other (specify)_____ ( ) ( ) will the work introduce a new hazard (specify)_____

WORK REQUIREMENTS

yes no yes no n/a ( ) ( ) fall protection/retrieval equipt. ( ) ( ) ( ) isolation of lines/systems ( ) lifeline ( ) harness ( ) tripod ( ) ( ) ( ) lockout-tagout (elect./mech.) ( ) other_____ ( ) ( ) ( ) entry with energized cables ( ) ( ) ground fault circuit interrupter (GFCI) ( ) ( ) ( ) cleaning, purging, inerting ( ) ( ) explosion-proof equipt./non-sparking tools ( ) ( ) ( ) space well ventilated (describe) ( ) ( ) lighting (specify)_____ _____ ( ) ( ) fire extinguisher(s) ( ) ( ) ( ) standby attendant assigned ( ) ( ) barricades, barriers, cones, etc. ( ) ( ) ( ) air monitor person assigned ( ) ( ) ladders ( ) ( ) ( ) pre-entry air monitoring PERSONAL PROTECTIVE EQUIPMENT ( ) ( ) ( ) continuous air monitoring ( ) ( ) gloves_____ ( ) ( ) ( ) rescue team notified of entry ( ) ( ) hardhat ( ) ( ) ( ) entry team trained ( ) ( ) foot protection_____ ( ) ( ) ( ) pre-entry briefing ( ) ( ) hearing protection ( ) ( ) ( ) other permits_____ ( ) ( ) eye/face_____ ( ) ( ) ( ) communications for entry team ( ) ( ) respirators ( ) radio ( ) visual ( ) other ( ) supplied air (specify)_____ _____ ( ) air purifying (specify)_____ ( ) ( ) ( ) emergency communications ( ) ( ) chemical protective clothing_____ ( ) radio ( ) phone ( ) other_____ ( ) ( ) radiation protective clothing_____ _____ ( ) ( ) other (specify)_____ ( ) ( ) ( ) other (specify)_____ _____ _____

SPECIAL INSTRUCTIONS:_____

_____

AIR MONITORING

Monitoring required: ( ) prior to entry ( ) prior to re-entry after breaks ( ) continuously during entry ( ) other_____ Type of monitoring required: ( ) oxygen ( ) flammability ( ) CO ( ) H2S ( ) other_____( ) other_____ Personal monitoring required: ( ) yes ( ) no specify_____ Type of air monitor:_____Cal. Date:_____Cal. Due:_____ Fresh air field check of air monitor: ( ) OK ( ) NOT OK (explain)_____

AIR MONITORING RESULTS

Date time Oxygen (19.5-23.5%) LEL (<10%) Toxics (specific) 1 Toxics (specific) 2 Comments

_____
_____
_____

Toxics 1 specify agent and acceptable entry condition:_____ Toxics 2 specify agent and acceptable entry condition:_____

CONFINED SPACE WORKER(S) SIGNOFF:_____

_____
_____

STANDBY ATTENDANT(S) SIGNOFF:_____

_____
_____

ENTRY SUPERVISOR(S) SIGNOFF:_____

_____
_____

SUPERVISOR(s) SIGNOFF:_____
_____
_____

MH CONSTRUCTION FALL CONFINED SPACE ENTRY PLAN SIGN-OFF LOG

NAME (print) SIGNATURE SOCIAL SECURITY # DATE

_____

MISH 16 — Confined Space Entry

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https://www.mhc-gc.com/resources/safety-manual/section/confined-space-entry